Financial Fraud analysis and detection

🛡️ Financial Fraud Detection System An End-to-End Machine Learning Solution for Real-Time Risk Assessment

📋 Project Summary In the modern financial landscape, fraudulent transactions account for billions in annual losses. This project provides a scalable, automated solution to identify suspicious activity within massive datasets.

Using a dataset of over 6.3 million transactions, I developed a machine learning model that successfully identifies fraud with 94% accuracy. The project moves beyond static analysis by providing a live, interactive dashboard where stakeholders can input transaction data to receive an instant risk assessment.

🚀 Key Features Predictive Intelligence: Leverages a Logistic Regression model optimized for "needle-in-a-haystack" detection in imbalanced data.

Automated Pipeline: Integrated Scikit-Learn pipelines ensure that raw data is scaled and encoded instantly without manual intervention.

Stakeholder Dashboard: A Streamlit-powered interface designed for non-technical users to verify transaction integrity on the fly.

Scalability: Tested on millions of records to ensure high performance and low latency during inference.

🛠️ Tech Stack Language: Python 3.11

Modeling: Scikit-Learn (Pipelines, Logistic Regression, Standard Scaler)

Data Analysis: Pandas, NumPy

Visualization: Matplotlib, Seaborn

App Framework: Streamlit

Deployment: Streamlit Cloud / GitHub

📊 Methodology & Performance:

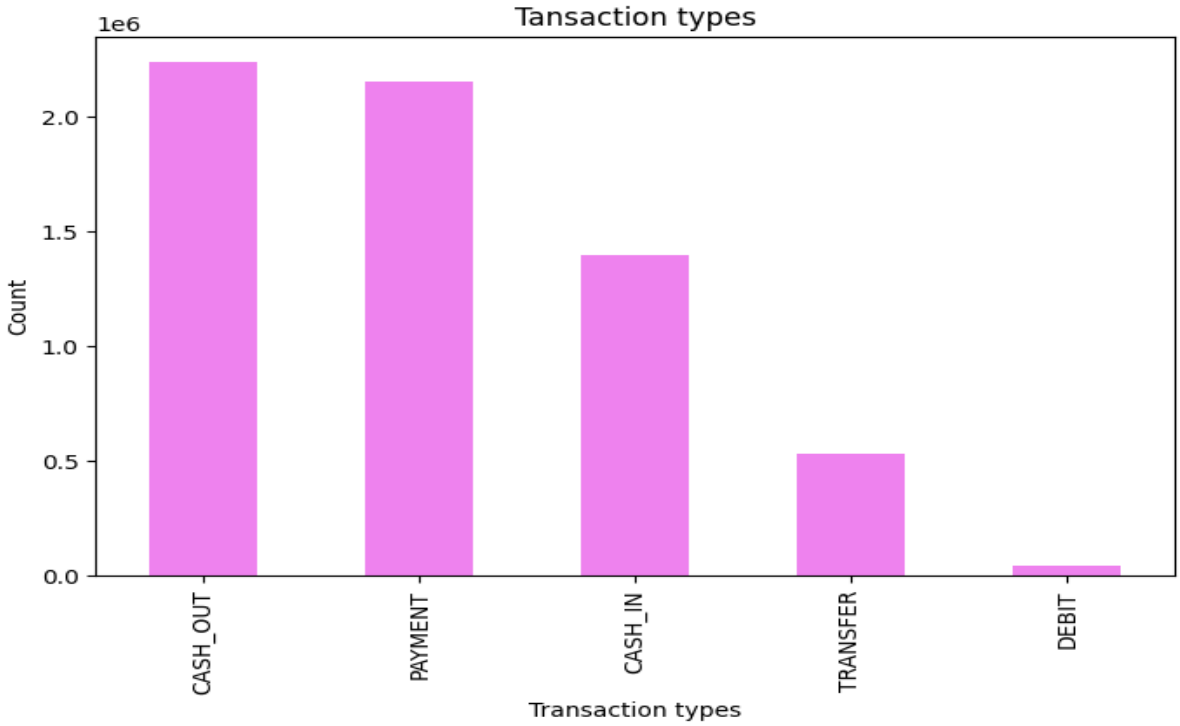
Identifying fraud is challenging because 99.8% of transactions are legitimate. To solve this, I implemented:

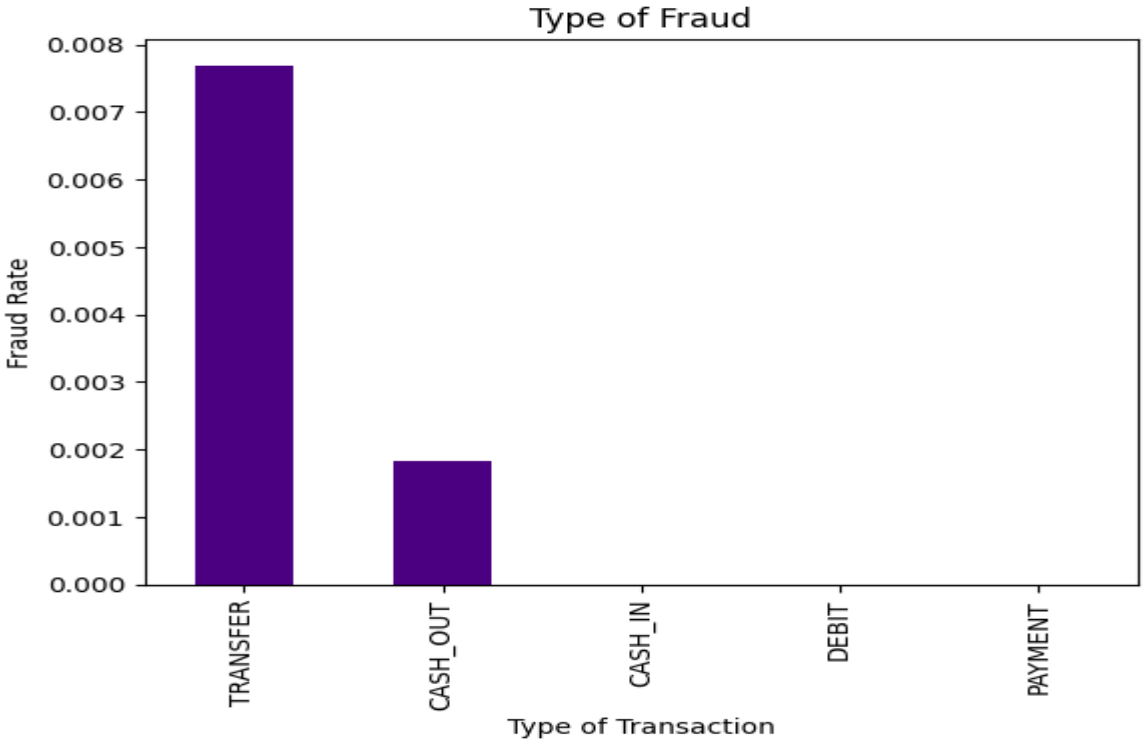
Class Balancing: Adjusted model weights to prioritize catching the 13% of fraudulent cases.

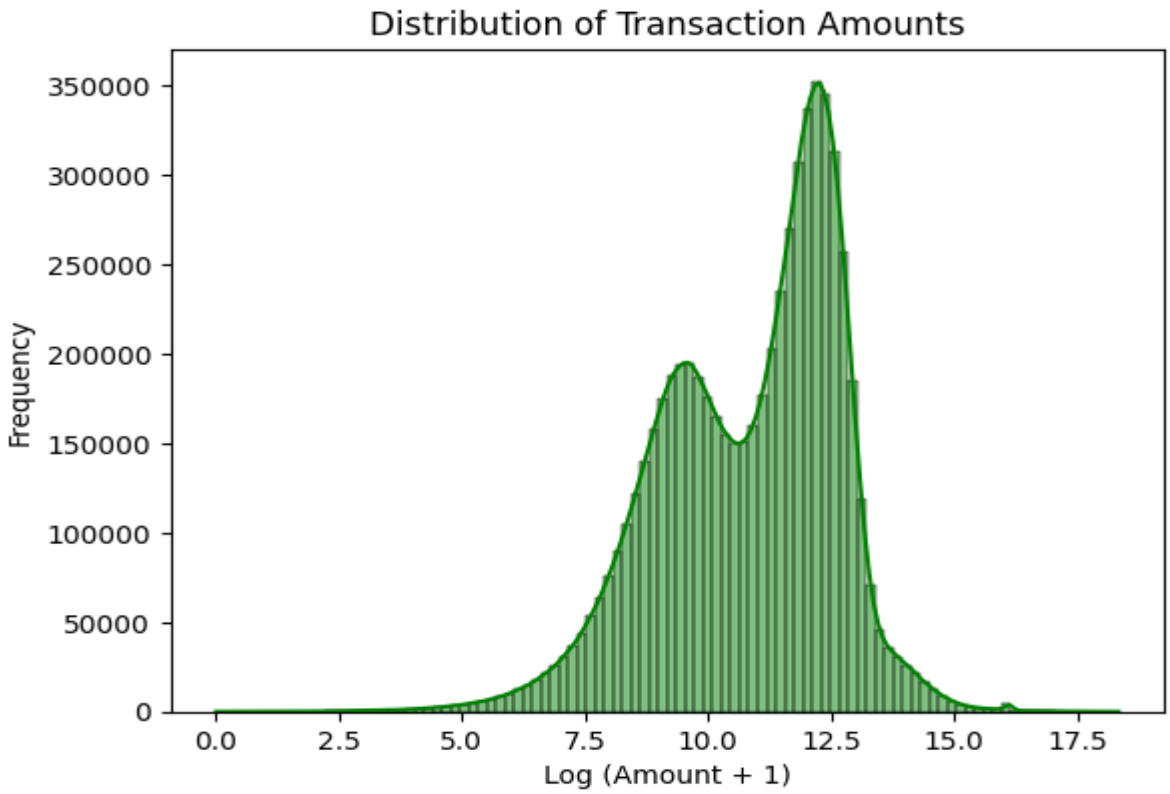
Feature Selection: Analyzed transaction types, focusing on 'Transfer' and 'Cash Out' activities which showed the highest correlation with fraud.

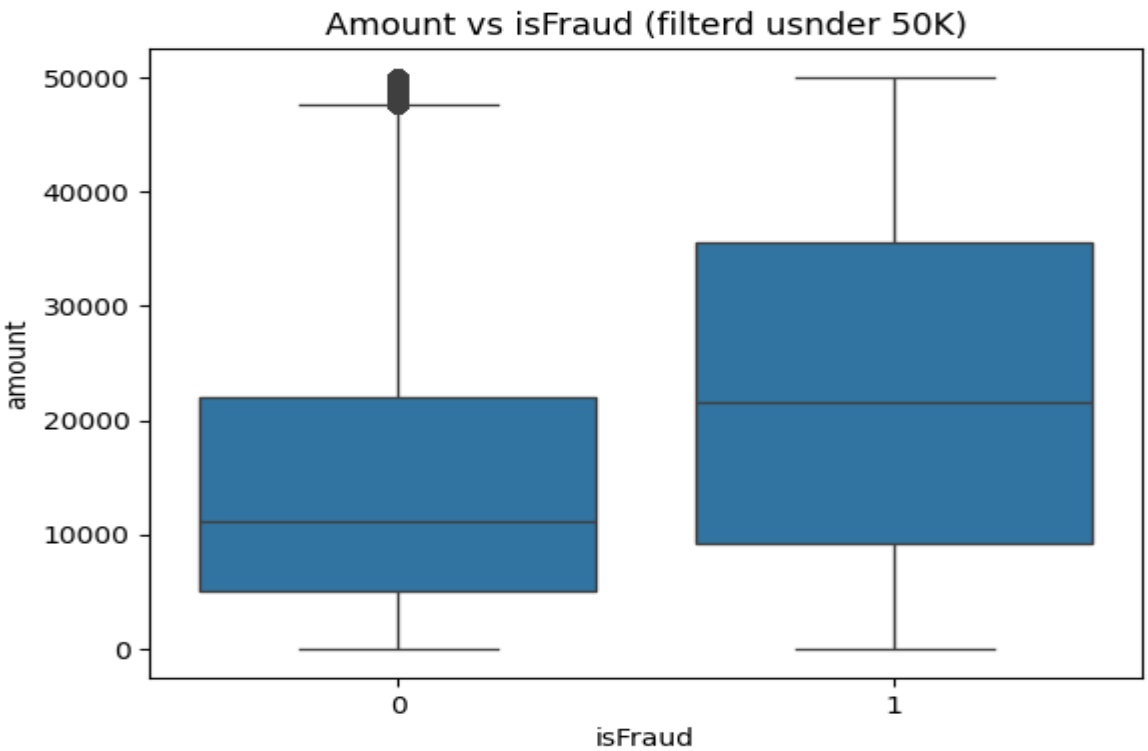
Performance: The model maintains a high accuracy rate while minimizing "False Positives" to avoid disrupting legitimate customer experiences.

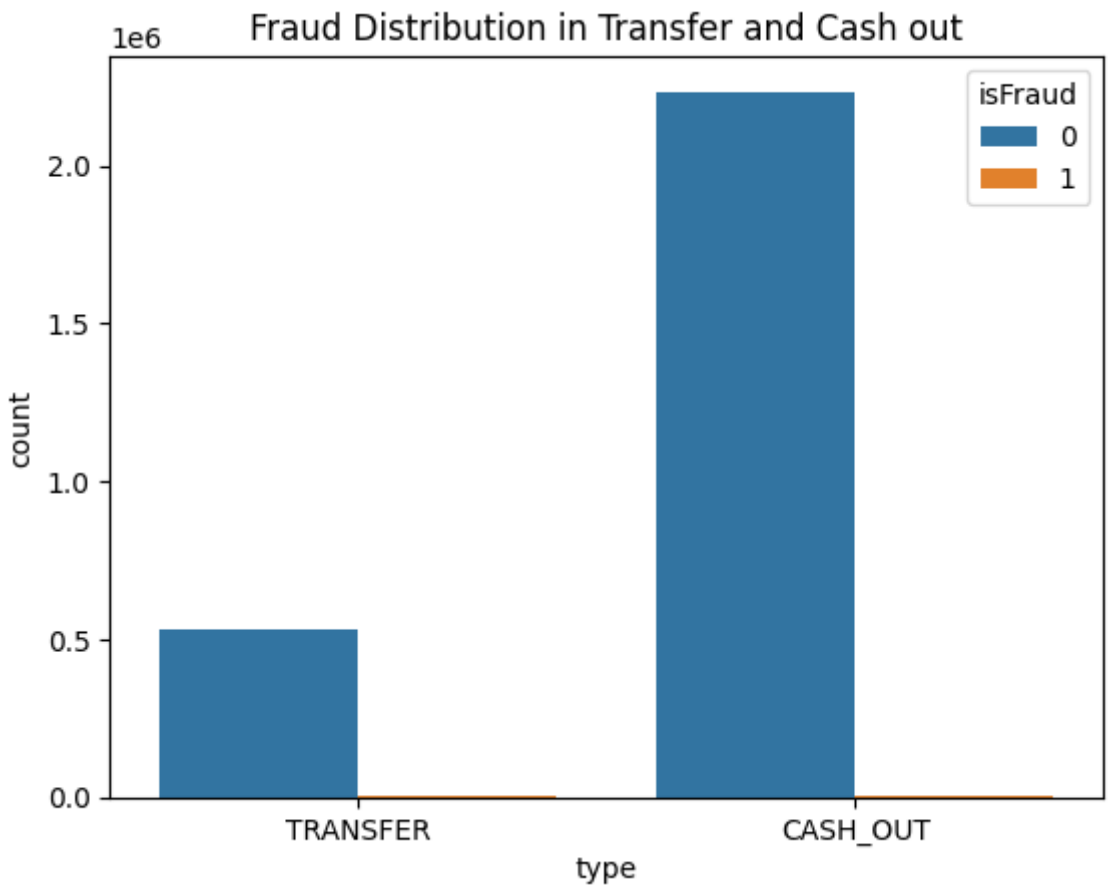
Data visualizations:


Tansaction types
1e6
2.0
1.5
1.0
0.5
0.0
Count
CASH_OUT
PAYMENT
CASH_IN
TRANSFER
DEBIT
Transaction types

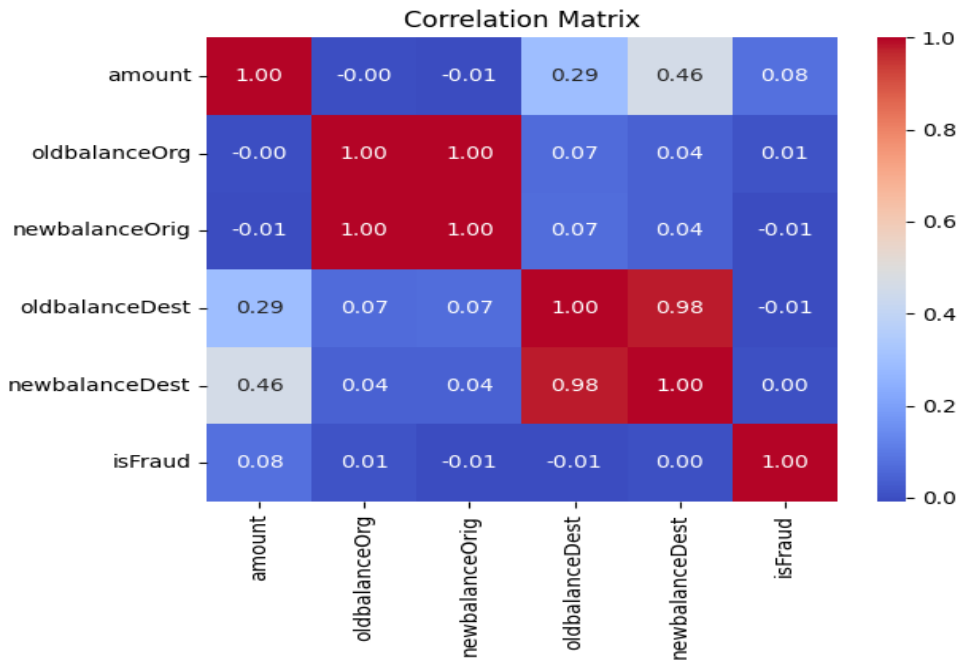

Type of Fraud
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0.000
Fraud Rate
TRANSFER
CASH_OUT
CASH_IN
DEBIT
PAYMENT
Type of Transaction


Distribution of Transaction Amounts
Frequency
350000
300000
250000
200000
150000
100000
50000
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Log (Amount + 1)

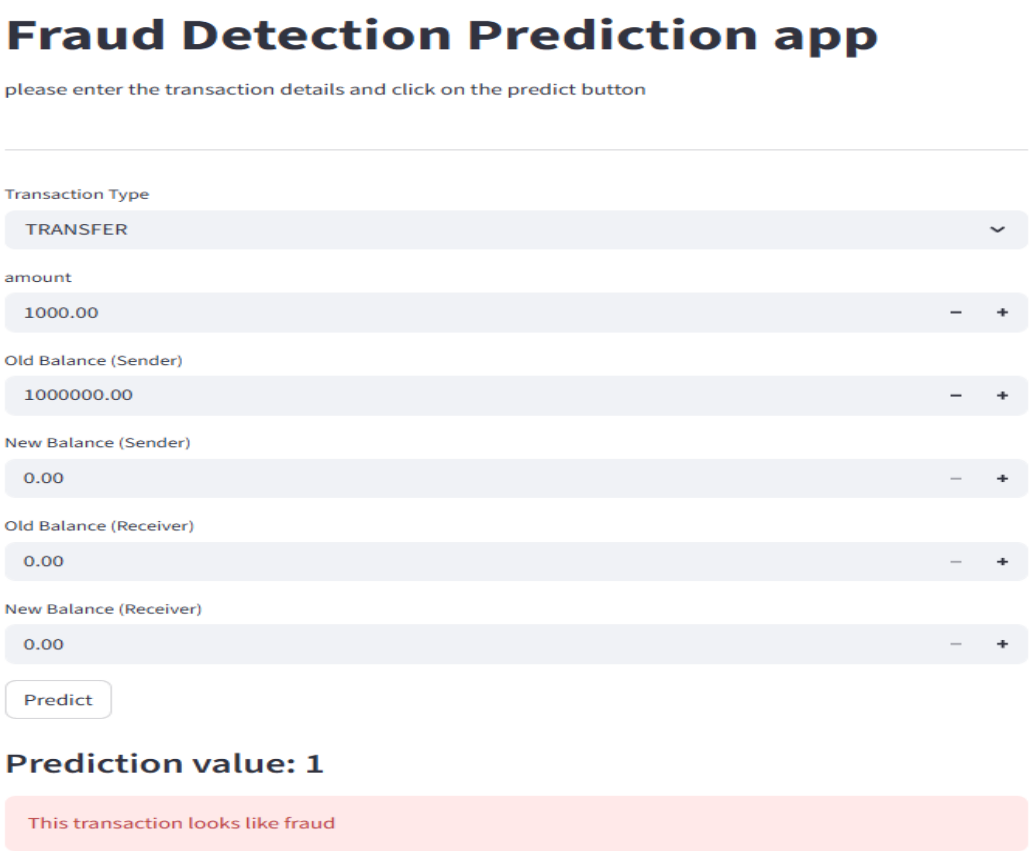

Amount vs isFraud (filterd usnder 50K)
amount
50000
40000
30000
20000
10000
0
0
1
isFraud



Select the Transaction Type.



Enter the Transaction Amount and the Account Balances.

Click Predict to see the system's risk assessment.

Note: A prediction of "1" indicates a high-risk transaction that requires manual review, while "0" indicates a low-risk, legitimate transaction.